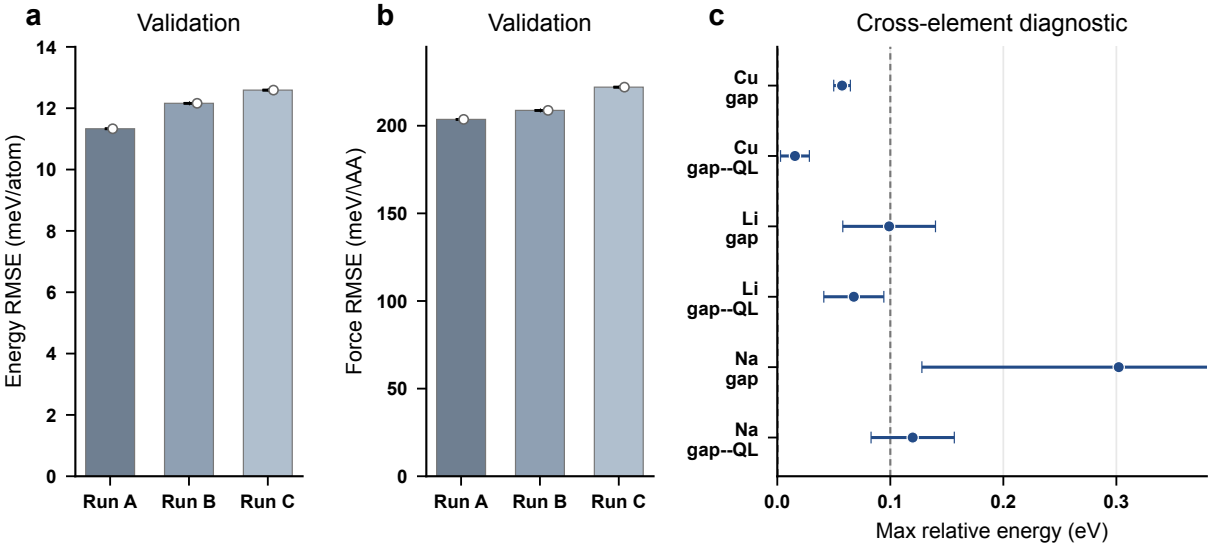Three-checkpoint broad-roster reproducibility and cross-element stress test